

PHASE 1

Data Collection & Data Engineering

2016 – 2025 NFL Seasons

Bearlytics is a portfolio-grade analytics platform built on ten years of Chicago Bears and NFL data. Phase 1 establishes the automated data pipeline, project architecture, and master datasets that power all subsequent analysis — replicating how a professional business intelligence team structures a reporting solution from the ground up.

Step 1 — Project Architecture

A standardized directory structure was designed before any data collection began, separating every stage of the analytics workflow and ensuring the project scales cleanly across phases.

```
Bearlytics/  
  01_raw_data/  
    team_stats/ game_logs/ player_stats/  
    salary_data/ league_stats/ logos/  
    cache/  
    02_excel_cleaning/ 03_clean_data/ 04_sql/  
    05_powerbi/ 06_screenshots/ 07_documentation/  
    08_README/ 09_scripts/
```

Raw data is further organized by dataset type rather than season, allowing individual data sources to be refreshed independently.

Step 2 — Automated Data Collection

Python automation scripts retrieve publicly available NFL data covering the 2016–2025 seasons. Each script performs automated downloading, basic validation, and structured file generation without manual intervention. Five core datasets were collected:

• Chicago Bears Team Statistics

Season-level team performance: wins, losses, ties, points scored/allowed, point differential, passing/rushing production, offensive/defensive statistics, and turnover metrics.

• Chicago Bears Game Logs

Every regular season game from 2016–2025 including week, opponent, home/away, final result, points for/against, point differential, and running team record.

- **Chicago Bears Player Statistics**

Season-level player production: passing, rushing, receiving, touchdowns, total yards, games played, position, and team.

- **Chicago Bears Contract & Salary Data**

Historical contract information for roster investment analysis: contract value, guaranteed money, contract length, position, team, and player. Only publicly verifiable fields were retained.

- **NFL League Statistics**

League-wide team statistics for all 32 NFL franchises across the same ten-year window, providing the benchmarks required for league comparison dashboards.

Step 3 — Raw Data Preservation

All original downloaded datasets are preserved inside a dedicated cache directory `01_raw_data/cache/`. This supports full reproducibility — every cleaned dataset can be recreated without re-downloading source data, and future updates can be applied cleanly.

Step 4 — Data Pipeline Design

Rather than importing raw data directly into Power BI, a structured ETL pipeline separates extraction, transformation, and loading across distinct stages. This reduces manual rework and makes future updates efficient.



Step 5 — Automation

Python scripts handle the full data acquisition and processing workflow:

- Download publicly available historical datasets from source APIs
- Organize files into standardized directories automatically
- Generate season-level CSV files (one per season, per dataset)
- Build consolidated master datasets spanning 2016–2025
- Handle missing directories without manual setup
- Preserve original source files in the cache directory
- Provide reusable scripts for future season updates

Phase 1 Deliverables

At the completion of Phase 1, the project contains five master datasets, fifty seasonal CSV files (ten per dataset), automation scripts, and a complete analytics project directory structure.

Dataset	File	Records Span
Team Statistics	bears_team_stats_2016_2025.csv	10 seasons
Game Logs	bears_game_logs_2016_2025.csv	10 seasons
Player Statistics	bears_player_stats_2016_2025.csv	10 seasons
Salary / Contracts	bears_salary_data_2016_2025.csv	10 seasons
NFL League Stats	nfl_league_stats_2016_2025.csv	10 seasons × 32 teams

Outcome

Phase 1 transitions Bearlytics from an empty project directory into a structured analytics platform with an automated data pipeline spanning ten NFL seasons. The project now has reproducible data acquisition scripts, standardized datasets, and a scalable foundation for SQL analysis and Power BI visualization — closer to how business intelligence teams build reporting solutions in industry than a dashboard assembled from manually downloaded spreadsheets.

Project Roadmap



Highlighted phase is complete. Remaining phases cover data modeling, SQL analysis, Power BI DAX, executive dashboard development, and GitHub portfolio presentation.

